

6 Marine Metal Mining

Two of the biggest crisis facing humanity are those of our dwindling potable water supply and declining natural resources. According to researchers, the ocean might hold the key to these problems. Developments in marine mining technology using catalyst assisted chemical engineering techniques have shown promise. This process not only purifies seawater to create a drinkable water



supply but also allows us to harvest minerals from the leftover brine waste. The brine waste created can be used in offsetting the high cost of making potable water as it will help us mine elements like lithium, magnesium, uranium and other high value minerals in one singular process.

8 Deep Learning Machines

The first step to creating an artificial intelligence is to find a powerhouse of computing and data storage. By employing a vast resource database of information, a machine can be programmed to learn independently by understanding language, and then making inferences and decisions. In today's internet driven world, companies like Google already hold the vast amount of computing power and storage needed and by partnering with researchers like Ray Kurzweil we already have an ongoing project that seeks to create a "deep learning machine". These software driven machines are structured to mimic the activity of neurons in the human neocortex to learn patterns in all forms of digital input to slowly become autonomous.



10 Screenless Display

Display screens are changing from objects that are looked at, to being a naturally organic extension of an users field of view. Innovations like Google Glass and Oculus Rift – use augmented reality and realistically created digital worlds respectively and are changing the way we interact with the world. The next wave of innovation in three dimensional holographic projection displays will further blur the line between our real and digital lives making screen-less displays our window to the future.



12 RNA-based Therapeutics

Innovations in the development of pharmaceuticals and other treatment technologies is another aspect of medical sciences that is undergoing a sea of change. Scientists are dedicated to unravelling the power of the RNA component biological specimens so they can create ever more powerful medical drugs and treatments. Targeted RNA therapies are being tested to preemptively disrupt the occurrence of diseases by blocking their messenger signals – this could lead to cures and treatment for cancer, tumours, heart disease and hepatitis virus.



7 Augmented Humanity

The highest potential of technological development is always its eventual invisibility. From electricity to mobile devices, we take the greatest modern technologies for granted due to their seamless integration into our lives. The next step in this evolution is for humans themselves to become carriers of technology which aid in our day to day lives. By "hacking" the mysteries of the human body scientists are looking for ways to create technologies such as bionic implants and exoskeletons that will change the way we live forever.



9 Grid-scale Electricity Storage

The requirement of energy and storage is the biggest issue in the world. To tackle this crisis, the development of grid-scale electricity storage has taken off with live projects around the world in places like Japan, Spain, United States and Europe - many of them showing immense potential. Using diverse technologies like high storage batteries, solar to thermal storage, molten salt storage and superconducting magnetic storage, innovators are not far from standardising a safe, reliable and efficient storage for the world's energy needs.



11 Human Microbiome Therapeutics

Medical sciences use cutting edge technologies to save lives everyday. But the use of human microbiome therapies can take it a step further by altering the very foundation of biological existence - at the gene level - to augment microbes within the human body in order to influence human biology. By harnessing the power over these microbes, scientists can



improve healthcare in a variety of ways such as treating diabetes, obesity, improve nutritional consumption and even defend against diseases such as cancer.

13 Quantified Self (Predictive Analytics)

Our real lives are intertwined with our digital lives - our medical data, our communication habits, our movement patterns and even our thoughts and feelings - form a quantified version of ourselves. By using powerful analytical algorithms, we can soon expect to find valuable results from our quantified selves. These millions of data points from our lives will soon provide us with a predictive analysis of all our needs in healthcare, work and lifestyle - to make our lives better. After all, that's what technology is primarily for.

